

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0536

Orebank 230 kV
April 23, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0536. The Interconnection Customer proposes a 98.6 MW wind facility interconnecting at the Orebank 230 kV substation in the Ohio area. The request was received on November 10, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,251,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Jones Chapel 500/230kV Auto
Contingency: Jones Chapel to Dekalb 500 kV L
Loading: 105.2% (Rating: 1,337 MVA)
DFAX: 5.7%
Pre-project loading: 104.7%
2. Facility: State St to Robison Park 230 kV
Contingency: Richland to Dekalb 500 kV
Loading: 102.9% (Rating: 1,124 MVA)
DFAX: 5.8%
Pre-project loading: 102.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jones Chapel 500 kV/230 kV Autotransformer	Replace transformer with higher-rated unit	\$38,202,000	5% (\$2,076,000)
State St to Robison Park 230 kV	Reconductor line	\$3,306,000	5% (\$175,000)
	Total allocated to project		\$2,251,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.